

2	(a) force <u>per unit positive</u> charge (on a small test charge)	B1	[1]
	(b) field strength = $(210 / \{1.5 \times 10^{-2}\}) = 1.4 \times 10^4 \text{ N C}^{-1}$	A1	[1]
	(c) (i) acceleration = Eq / m	C1	
	$= (1.4 \times 10^4 \times 1.6 \times 10^{-19}) / (9.1 \times 10^{-31})$	C1	
	$= 2.5 \times 10^{15} \text{ m s}^{-2} \quad (2.46 \times 10^{15})$	A1	
	towards positive plate / upwards (and normal to plate)	B1	[4]
	(ii) time = $2.4 \times 10^{-9} \text{ s}$	A1	[1]
	(d) <i>either</i> vertical displacement after acceleration for $2.4 \times 10^{-9} \text{ s}$		
	$= \frac{1}{2} \times 2.46 \times 10^{15} \times (2.4 \times 10^{-9})^2$	C1	
	$= 7.1 \times 10^{-3} \text{ m}$	A1	
	(0.71 cm < 0.75 cm and) so will pass between plates	A1	[3]
	<i>i.e. valid conclusion based on a numerical value</i>		
	or $0.75 \times 10^{-2} = \frac{1}{2} \times 2.46 \times 10^{15} \times t^2$	(C1)	
	t is time to travel 'half-way across' plates = $2.47 \times 10^{-9} \text{ s}$	(A1)	
	(2.4 ns < 2.47 ns) so will pass between plates	(A1)	
	<i>i.e. valid conclusion based on a numerical value</i>		

3	(a) mass / volume (ratio idea essential)	B1	[1]
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